

Finite edge graphs cannot realize every real value

Disproof of Conjecture 000000008883

ziangni-sys

4 October 2026

The finite-realization clause

Both originals assert that every real surreal can be realized by a finite edge graph in blue-red Hackenbush. We disprove this clause by a cardinality obstruction that applies to every possible deterministic value rule on finite grounded colored graphs. It does not require a formula for Hackenbush values or an analysis of infinite positions.

A blue-red Hackenbush position is an abstract colored graph with a ground set: a player removes an edge of the appropriate color, after which edges disconnected from ground disappear. Only edge incidence, colors and grounding affect the game. Relabeling vertices or edges and redrawing the same graph do not change the legal moves or the game value. Distinct real numbers represent distinct real surreals, so realizing all real surreals would require realizing all real numbers as values of these positions.

Exhaustive finite descriptions

Allow a grounded multigraph with any finite edge set E , any vertex set V , endpoint maps $s, t : E \rightarrow V$, edge colors $b : E \rightarrow \{0, 1\}$, and ground flags $a : V \rightarrow \{0, 1\}$. This includes parallel edges and loops, so we do not obtain a bound by restricting to simple graphs. The vertex set itself need not be finite. Its incident support is

$$S = s(E) \cup t(E),$$

a finite set. Restricting to S retains every edge and both endpoints, all edge colors, and the ground status of every incident vertex. Vertices outside S have no incident edge and can never provide a move or a connection between edges. Removing them changes no play. This also handles the source's wording "finite edges" without silently assuming a finite vertex set.

Enumerate S and E as $\{0, \dots, v-1\}$ and $\{0, \dots, e-1\}$. The normalized graph becomes a finite description consisting of

$$s, t : \text{Fin}(e) \rightarrow \text{Fin}(v), \quad b : \text{Fin}(e) \rightarrow \{0, 1\}, \quad a : \text{Fin}(v) \rightarrow \{0, 1\}.$$

For fixed v, e there are finitely many such descriptions ($v^{2e}2^{e+v}$, with the usual empty-function conventions). Taking the union over all natural v, e gives a countable type D . The labeling is exhaustive: any finite-edge position first normalizes as above and then has one of these descriptions, with all incidence, color and ground data preserved. Any disconnected or initially unplayable description only enlarges this countable set.

No value rule can cover the reals

The real numbers are uncountable. Hence no function $D \rightarrow \mathbb{R}$ is surjective. Even if some descriptions do not have a real value, the subtype of descriptions that do is still countable, so no partial real-value rule can be surjective either. Equivalently, for any relation $R(d, r)$ in which a description has at most one real value, the assertion $\forall r \in \mathbb{R} \exists d \in D R(d, r)$ is false. A Hackenbush position has one game value, and isomorphic descriptions have the same value. Thus the conjectured realization of every real surreal by finite edge graphs would give exactly this impossible coverage. The finite-realization clause is false, irrespective of the separate claimed correspondence for infinite Hackenbush.

Formal model and completeness

Lean defines actual graph data `Graph V E` by endpoint maps, Boolean edge colors and Boolean ground flags. The edge type is independent of endpoints, so multiple edges are genuinely allowed. `Description v e` contains these same maps over actual finite index types. The full description type is a dependent sum over all natural vertex and edge counts, not a bounded collection. Mathlib’s countability handler encodes each fixed-size description, and countability instances cover the dependent sum.

`IsRelabelling` explicitly requires both endpoint equations, the color equation for every edge, and the ground equation for every vertex, under actual vertex and edge bijections.

`describe_preserves_everything` constructs finite index representations for arbitrary finite vertex and edge types and proves all these equations. No completeness of labeling is assumed.

To cover arbitrary vertex sets, `support` is the actual finite union of endpoint images. The source/target support theorems prove that every endpoint lies there. `normalize` uses the actual subtype of incident vertices and retains the same edge type and colors. `normalization_preserves_incidence` proves all endpoint, color and ground-flag equalities after forgetting the subtype labels.

`exhaustive_finite_edge_representation` combines this normalization with finite enumeration. It quantifies over every vertex type and every finite edge type and produces a finite description plus actual vertex/edge equivalences satisfying the full relabeling condition.

Formal cardinal contradiction

The type `FiniteDescription` is proved countable. `no_value_assignment_realizes_all_reals` proves that *every* function from that type to the actual real numbers is nonsurjective. `omitted_real` gives an omitted real for each such function. `no_partial_value_assignment` covers any subtype of eligible descriptions, so real-valuedness of every position is not a hidden premise.

The theorem `no_functional_realization_covers_all_reals` quantifies over any realization relation with at most one real value per description and negates coverage of all reals. Its proof selects the unique value on eligible descriptions; full coverage would make that selected map surjective, contradicting the preceding theorem. This is a universal necessary-condition disproof: no special Hackenbush evaluation table, dyadic-value theorem, or desired nonsurjectivity is assumed. The graph semantics supplies only the ordinary uniqueness of a position’s value and invariance under relabeling/removal of vertices incident to no edge, justified above.

Verification

The project uses Lean 4.19.0 with pinned Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. Axiom audits of exhaustive representation, nonsurjectivity, the omitted real theorem and the realization-relation obstruction use only `propext`, `Classical.choice` and `Quot.sound`. No admitted proofs, additional axioms, native evaluation shortcuts or auxiliary computation are used. The complete graph model includes all finite sizes and arbitrary grounding; labeling artifacts or drawing geometry cannot add real-valued game information.